

Advanced Dynamics — Equation Sheet

Kinematics · Newton's Laws · Energy & Momentum · Generalized Coordinates & Constraints · D'Alembert's Principle

Notation

- $N, A, B, \dots$ — reference frames (e.g. N = fixed/inertial)
- $\hat{n}_i, \hat{a}_i, \hat{b}_i$ — right-handed unit-vector basis fixed in N, A, B
- $\vec{r}_{P/O}$ — position of point P relative to point O ; $\vec{r} \equiv {}^N\vec{r}$ unless noted otherwise
- ${}^N\vec{v}_P, {}^N\vec{a}_P$ — velocity/acceleration of P as seen from N
- $\vec{\omega}^{A/N}$ — angular velocity of frame A relative to N
- $\vec{\alpha}^{A/N} = N \frac{d}{dt} \vec{\omega}^{A/N}$ — angular acceleration of A relative to N
- $N \frac{d}{dt}(\cdot)$ — time derivative taken by an observer fixed in N
- $[R]^{B/A}$ — rotation (direction cosine) matrix: $\{\hat{b}\} = [R]^{B/A} \{\hat{a}\}$
- $q_1, \dots, q_N$ — generalized coordinates; $\tilde{\gamma}_{ij} = \partial \vec{r}_i / \partial q_j$ — partial velocities
- $\vec{P} = m\vec{v}$ — linear momentum; $\vec{H}_O = \vec{r} \times \vec{P}$ — angular momentum about O

Rotations & Angular Velocity

Rotation matrices

Orthogonal, so $[R]^{-1} = [R]^T$; composing rotations through an intermediate frame A :

$$[R]^{B/N} = [R]^{B/A} [R]^{A/N}$$

Angular velocity addition (chain rule)

Valid for any chain of intermediate frames:

$$\vec{\omega}^{B/N} = \vec{\omega}^{B/A} + \vec{\omega}^{A/N}, \quad \vec{\omega}^{C/N} = \vec{\omega}^{C/B} + \vec{\omega}^{B/A} + \vec{\omega}^{A/N}$$

Note the reversal/cancellation identities:

$$\vec{\omega}^{A/N} = -\vec{\omega}^{N/A}, \quad \vec{\omega}^{A/A} = \vec{0}$$

Derivative of a rotating unit vector

If $\hat{b}_i$ is fixed in B (constant length/orientation relative to B):

$$N \frac{d}{dt} \hat{b}_i = \vec{\omega}^{B/N} \times \hat{b}_i$$

2D rotation example (N, P frames offset by α)

$$\hat{n}_1 = \hat{p}_1 \cos \alpha + \hat{p}_2 \sin \alpha, \quad \hat{n}_2 = -\hat{p}_1 \sin \alpha + \hat{p}_2 \cos \alpha$$

$$\begin{Bmatrix} \hat{n}_1 \\ \hat{n}_2 \end{Bmatrix} = [R]^{N/P} \begin{Bmatrix} \hat{p}_1 \\ \hat{p}_2 \end{Bmatrix}, \quad [R]^{N/P} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

(written ${}^N R_P$ in lecture)

$$[R]^{P/N} = ([R]^{N/P})^{-1} = ([R]^{N/P})^T = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

Vector Differentiation Formula (VDF)

Also called the *transport theorem*: for *any* vector $\vec{q}$ (not necessarily fixed in either frame), relating the rate of change seen in N to the rate of change seen in a frame A rotating at $\vec{\omega}^{A/N}$:

$$N \frac{d}{dt} \vec{q} = A \frac{d}{dt} \vec{q} + \vec{\omega}^{A/N} \times \vec{q}$$

Corollary: if $\vec{q}$ is fixed in A (i.e. $A \frac{d}{dt} \vec{q} = 0$), this reduces to

$N \frac{d}{dt} \vec{q} = \vec{\omega}^{A/N} \times \vec{q}$ (previous box). This is the tool used to differentiate a vector expressed in a rotating basis one component at a time.

Position Vector Addition (Setup for Moving Frames)

For two points P and O' both located relative to a common origin O (equivalently $\vec{r} = \vec{R} + \vec{r}'$, with $\vec{R} = \vec{r}_{O'/O}$, $\vec{r}' = \vec{r}_{P/O'}$), position vectors add/subtract exactly like any vectors — the basic relation used to move between frames whose origins differ:

$$\vec{r}_{P/O} = \vec{r}_{P/O'} + \vec{r}_{O'/O} \quad \Longleftrightarrow \quad \vec{r}_{P/O'} = \vec{r}_{P/O} - \vec{r}_{O'/O}$$

Chained over several intermediate points: $\vec{r}_{P/O} = \vec{r}_{P/B} + \vec{r}_{B/A} + \vec{r}_{A/O}$. Applying the VDF to this relation term-by-term is exactly how the velocity/acceleration relations below are derived.

Velocity: Translating & Rotating Frames

Let O' be the (moving) origin of frame A , which translates *and* rotates ($\vec{\omega}^{A/N}$) relative to N , and let P be an arbitrary point:

$$N \vec{v}_P = N \vec{v}_{O'} + A \vec{v}_P + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$$

- $N \vec{v}_{O'}$ — translational velocity of the moving origin
- $A \vec{v}_P = A \frac{d}{dt} \vec{r}_{P/O'}$ — velocity of P relative to A (as seen by an observer riding on A)
- $\vec{\omega}^{A/N} \times \vec{r}_{P/O'}$ — velocity contribution from A 's rotation

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): $N \vec{v}_P = N \vec{v}_{O'} + A \vec{v}_P$
- P fixed in A ($A \vec{v}_P = 0$, rigid-body point): $N \vec{v}_P = N \vec{v}_{O'} + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$
- Common origin ($O' = O$, $N \vec{v}_{O'} = 0$): reduces to the transport theorem applied to $\vec{r}_{P/O}$

Acceleration: Translating & Rotating Frames

Differentiating the velocity relation again (the “five-term” / Coriolis theorem):

$$N \vec{a}_P = N \vec{a}_{O'} + A \vec{a}_P + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'} + \vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'}) + 2 \vec{\omega}^{A/N} \times A \vec{v}_P$$

- $N \vec{a}_{O'}$ — translational acceleration of the moving origin
- $A \vec{a}_P = A \frac{d}{dt} A \vec{v}_P$ — acceleration of P relative to A
- $\vec{\alpha}^{A/N} \times \vec{r}_{P/O'}$ — Euler (angular acceleration) term
- $\vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$ — centripetal term
- $2 \vec{\omega}^{A/N} \times A \vec{v}_P$ — Coriolis term

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): $N \vec{a}_P = N \vec{a}_{O'} + A \vec{a}_P$
- P fixed in A ($A \vec{v}_P = A \vec{a}_P = 0$, rigid-body point):

$$N \vec{a}_P = N \vec{a}_{O'} + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'} + \vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$$
- P moving in A but O' fixed in N (common origin, no translation): drop the $N \vec{a}_{O'}$ term — this is the form used for rotating-basis problems (e.g. spherical pendulum, rotating tube)

Newton's Laws of Motion

$$\text{1st: } \sum \vec{F} = 0 \Rightarrow N \ddot{\vec{r}} = \text{const} \quad \text{2nd: } \sum \vec{F} = m N \ddot{\vec{r}}$$

$$\text{3rd: } \vec{F}_{12} = -\vec{F}_{21} \quad (\text{action–reaction pairs})$$

Newton's 2nd law **requires** the acceleration relative to an inertial (Newtonian) frame N — never a rotating/accelerating frame directly.

Work & Energy

Kinetic & potential energy

$$T = \frac{1}{2} m |\dot{\vec{r}}|^2 = \frac{1}{2} m \dot{\vec{r}} \cdot \dot{\vec{r}}, \quad V = V(\vec{r})$$

$$\vec{F} = -\nabla V(\vec{r}) \quad (\text{conservative/“potential” force})$$

Common potentials

- Uniform gravity: $V = mgh$, $\vec{F} = -mg \hat{n}_3$
- Newtonian gravity: $V = -\frac{GmM}{r}$, $\vec{F} = \frac{GmM}{r^2} \hat{b}_1$ ($\hat{b}_1$: unit vector toward the attracting mass)
- Linear spring (stretch x): $V = \frac{1}{2} kx^2$, $\vec{F} = -kx \hat{n}_1$

Work–energy theorem

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}, \quad T_2 - T_1 = W$$

All forces conservative ($\vec{F} = -\nabla V$):

$$T_1 + V_1 = T_2 + V_2 \quad (\text{energy conserved})$$

Only holds when *every* force doing work is conservative — friction or other non-conservative applied forces break it.

Momentum

Linear momentum

$$\vec{P} = m\dot{\vec{r}}, \quad \dot{\vec{P}} = m\ddot{\vec{r}} = \sum \vec{F}$$

If $\sum \vec{F} = \vec{0}$, $\vec{P}$ is conserved.

Angular momentum (about inertial fixed point O)

$$\vec{H}_O = \vec{r} \times \vec{P}, \quad \dot{\vec{H}}_O = \vec{r} \times \sum \vec{F} = \sum \vec{M}_O$$

“Rotational Newton’s 2nd law.” If $\sum \vec{M}_O = \vec{0}$, $\vec{H}_O$ is conserved.

Generalized Coordinates

- **Generalized coordinates** $q_1, \dots, q_N$: minimal (or redundant) geometric variables (angles/distances) describing the configuration of a system
- **Configuration space**: the set of all configurations the system can take on, parametrized by the q_j
- A system may be described by a **redundant set** (more coordinates than DOF, related by constraint equations) or a **minimal set** (one coordinate per DOF, no constraints left)

$$n_{\text{DOF}} = \#(\text{gen. coords}) - \#(\text{indep. constraints})$$

Always reduce to the minimal set before deriving equations of motion.

Holonomic Constraints

Position-based: with N generalized coordinates and M constraints ($j = 1, \dots, M$):

$$\phi_j(q_1, \dots, q_N) = \phi_j(\vec{q}) = 0, \quad \text{or, if time-dependent, } \phi_j(\vec{q}, t) = 0$$

- **Scleronomic**: no explicit t — time-independent constraint
- **Rheonomic**: explicit t dependence — e.g. a prescribed moving support
- Each independent holonomic constraint removes exactly one DOF (box above)
- **Always rewritable as non-holonomic**: since $\phi_j \equiv 0 \Rightarrow \dot{\phi}_j \equiv 0$,

$$\dot{\phi}_j(\vec{q}, t) = \sum_{i=1}^N \frac{\partial \phi_j}{\partial q_i} \dot{q}_i + \frac{\partial \phi_j}{\partial t} = 0 \quad (a_{ji} \equiv \partial \phi_j / \partial q_i)$$

Non-Holonomic Constraints

Velocity-based: $\phi_j(\vec{q}, \dot{\vec{q}}, t) = 0$, usually taken *linear* in the velocities (Pfaffian form):

$$\phi_j = \sum_{i=1}^N a_{ji}(\vec{q}, t) \dot{q}_i + a_{jt}(\vec{q}, t) = 0$$

(no explicit t : $\phi_j = \sum_i a_{ji}(\vec{q}) \dot{q}_i = 0$)

- Every holonomic constraint can be written this way (box above); the **converse is false** — a Pfaffian constraint is genuinely non-holonomic unless it is exact/integrable (or made exact with an integrating factor; check via the Frobenius condition)
- Non-holonomic constraints reduce the number of independent *velocities* but **not** the dimension of the configuration space
- **Example — knife-edge/skate/wheel constraint**: contact-point velocity must be perpendicular to the blade, $\vec{v}_A \cdot \hat{n} = 0$ with $\hat{n} = (-\sin \theta, \cos \theta)$:

$$\phi = -\sin \theta \dot{x} + \cos \theta \dot{y} = 0$$

With $(q_1, q_2, q_3) = (x, y, \theta)$: $a_1 = -\sin q_3$, $a_2 = \cos q_3$, $a_3 = 0$. Cannot be integrated to $\phi(\vec{q}) = 0$ — genuinely non-holonomic.

D’Alembert’s Principle

Partial velocities

For particle i with position $\vec{r}_i(\vec{q}, t)$, the partial velocity w.r.t. generalized speed $\dot{q}_j$:

$$\vec{\gamma}_{ij} = \frac{\partial \vec{r}_i}{\partial q_j} = \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_j}$$

(equal because $\dot{\vec{r}}_i$ is linear in the $\dot{q}_j$ ’s — fastest computed by differentiating the already-known $\vec{r}_i$ directly w.r.t. $\dot{q}_j$.)

Equations of motion

For a minimal set $q_1, \dots, q_M$ and particles $i = 1, \dots, n$:

$$\sum_{i=1}^n (\vec{F}_{Ai} - m_i \ddot{\vec{r}}_i) \cdot \vec{\gamma}_{ij} = 0, \quad j = 1, \dots, M$$

- $\vec{F}_{Ai}$ — *applied* (impressed) forces on particle i only (gravity, springs, motors, ...)
- **Ideal constraint forces are excluded** (rod tension, normal forces, frictionless-contact forces) — they do no virtual work consistent with the constraints and drop out automatically
- One scalar equation per generalized coordinate — exactly M equations for M DOF
- Equivalently, with generalized force $Q_j = \sum_i \vec{F}_{Ai} \cdot \vec{\gamma}_{ij}$: $\sum_i m_i \ddot{\vec{r}}_i \cdot \vec{\gamma}_{ij} = Q_j$

Recipe

1. Choose a minimal set $q_1, \dots, q_M$ consistent with all holonomic constraints
2. Write $\vec{r}_i(\vec{q}, t)$ for every particle; differentiate for $\dot{\vec{r}}_i, \ddot{\vec{r}}_i$
3. Compute $\vec{\gamma}_{ij} = \partial \vec{r}_i / \partial q_j$ for every particle/coordinate pair
4. Identify applied forces $\vec{F}_{Ai}$ on each particle (*exclude* constraint forces)
5. For each j : $\sum_i (\vec{F}_{Ai} - m_i \ddot{\vec{r}}_i) \cdot \vec{\gamma}_{ij} = 0 \Rightarrow M$ equations of motion
6. For linear (spring/mass) systems, these assemble into $[M]\ddot{\vec{q}} + [K]\vec{q} = \vec{F}$